

TEST CASE MATRIX

EDUGENIE – AI-POWERED LEARNING ASSISTANT

ID | Feature | Input | Expected Output

TC01 | Startup | Start Uvicorn | Application starts

TC02 | UI | Open local URL | Main interface displayed

TC03 | Q&A | Academic question | AI-generated answer

TC04 | Explanation | Concept/topic | Simplified explanation

TC05 | Quiz | Topic/material | Multiple-choice quiz

TC06 | Summary | Long text | Concise summary

TC07 | Learning Path | Subject | Ordered learning path

TC08 | Configuration | Valid API environment | Gemini request can be configured

TC09 | Configuration | Missing API credential | Application should not expose a secret

TC10 | Local model | Valid explanation request | Local model generates explanation when resources are sufficient